(5.2.) 5. Find the characteristic polynomial and the eigenvalues of the matrix

$$\begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$$

Sol. Denote the given matrix by A. The characteristic polynomial is obtained as

$$\begin{aligned} p_A(\lambda) &= \det(A - \lambda I) \\ &= \begin{vmatrix} 2-\lambda & 1 \\ -1 & 4-\lambda \end{vmatrix} \\ &= (2-\lambda)(4-\lambda) - (-1) \\ &= \lambda^2 - 6\lambda + 9 = (\lambda - 3)^2. \end{aligned}$$

Set $p_A(\lambda) = 0$ to obtain the roots as $\lambda = 3$, which is the eigenvalue of A (with multiplicity 2).

10. Find the characteristic polynomial of the given matrix using expansion across a row or down a column. (Note: Finding the characteristic polynomial of a 3×3 matrix is not easy with just row operations, because the variable λ is involved.)

$$\begin{bmatrix} 0 & 3 & 1 \\ 3 & 0 & 2 \\ 1 & 2 & 0 \end{bmatrix}$$

Sol. The "expansion" is referring to the cofactor expansion in 3.1.
The characteristic polynomial is obtained by

$$\begin{aligned} \begin{bmatrix} -\lambda & 3 & 1 \\ 3 & -\lambda & 2 \\ 1 & 2 & -\lambda \end{bmatrix} &= -\lambda(-1)^{1+1} \begin{vmatrix} -\lambda & 2 \\ 2 & -\lambda \end{vmatrix} + 3(-1)^{1+2} \begin{vmatrix} 3 & 2 \\ 1 & -\lambda \end{vmatrix} + 1(-1)^{1+3} \begin{vmatrix} 3 & -\lambda \\ 1 & 2 \end{vmatrix} \\ &= -\lambda^3 + 14\lambda + 12. \end{aligned}$$

(5.3.) 11. Solved in discussion